

A Formal Proof of the Pythagorean Theorem

Andreas Mulard^{1,2,3*}, Yacine Hbada⁴, Léonie Gaztelumendi⁵,
Thomas Mairesse⁶, Romain Brouard^{4,7},
Madhav Cherupilil Sajeev^{1,3,8}, Alexis Pocquet^{1,3},
Fernando Leal^{3,9,10}, Florian Chivé^{1,3,8}, Lyam Goux⁴,
Tran Décaudin¹¹, Tarkus the Armadillo-Tank¹², Maike Päsler¹³,
Gabriel Berthel¹⁴, Max Lemoine¹⁵

¹Institut Polytechnique de Paris, Route de Saclay, Palaiseau, 91120, France.

²Conservatoire National des Arts et Métiers, 292 Rue Saint-Martin, Paris, 75003, France.

³MPRI Program, Ile de France, France.

⁴Université de Tours, 60 rue du Plat d'Étain, Tours, 37000, France.

⁵Université de Poitiers, 15 Rue de l'Hôtel Dieu, Poitiers, 86000, France.

⁶Université de Montpellier Paul-Valéry, Route de Mende, Montpellier, 34000, France.

⁷Polytech Tours, 64 Avenue Jean Portalis, Tours, 37200, France.

⁸Ecole Polytechnique, Route de Saclay, Palaiseau, 91120, France.

⁹École normale supérieure Paris-Saclay, 4 Avenue des Sciences, Gif-sur-Yvette, 91190, France.

¹⁰Université Paris-Saclay, 9 Rue Joliot Curie, Gif-sur-Yvette, 91190, France.

¹¹École Supérieure de Génie Informatique, 65 Rue Philippe-Laurent Roland, Lille, 59000, France.

¹²CMA Formation 41, 27 Rue André Boulle, Blois, 41000, France.

¹³Universität Bielefeld, Universitätsstraße 25, Bielefeld, 33615, Germany.

¹⁴Université de Montréal, 2900 Edouard Montpetit Boulevard, Montréal, Québec, Canada.

¹⁵Université de Mans, Avenue Olivier Messiaen, Le Mans, 72085, France.

*Corresponding author(s). E-mail(s): andreas.mulard@ip-paris.fr;

Contributing authors: yacine.hbada@etu.univ-tours.fr;
leonie.gaztel@gmail.com; madhav.cherupilil-sajeev@polytechnique.edu;
florian.chive@polytechnique.edu; lyam.goux@etu.univ-tours.fr;
tran.decaudin@proton.me; fernando.leal.sanchez@ens-paris-saclay.fr;

Abstract

The Pythagorean theorem is a foundational result typically encountered early in mathematical education, where it is accepted through diagrams, folklore, and social pressure. Despite being one of the best-known theorems in human history, it is rarely presented to students in a mechanized, machine-checkable form. We remedy this gap by providing a complete formal proof in the Rocq/Coq proof assistant. Our proof is intentionally minimal: we avoid geometry, avoid square roots, and reduce the statement to algebraic normalization in $\mathbb{R}$. The main technical contribution is the observation that `ring` is more consistent than most classrooms.

Keywords: formal proof, Pythagorean theorem, Rocq, Coq, ring tactic

1 Introduction

The Pythagorean theorem states that in a right triangle, the square of the hypotenuse equals the sum of the squares of the legs [1]. Most readers first encounter this theorem via a diagram containing at least one line that is not quite straight. They then proceed to use it for years without ever asking whether the proof was formally correct, or merely convincing¹.

Meanwhile, proof assistants have matured to a point where many deep results can be mechanized [2]. This naturally raises an urgent question: can we also mechanize the theorem that everyone already knows?

This paper answers “yes”, in the most literal and least surprising way possible.

2 Statement and Setup

Rather than formalizing Euclidean geometry, we place the right triangle on the coordinate axes. Let

$$O = (0, 0), \quad A = (a, 0), \quad B = (0, b),$$

with $a, b \in \mathbb{R}$. Then OA and OB are perpendicular, and the squared length of the segment AB is

$$AB^2 = (a - 0)^2 + (0 - b)^2 = a^2 + b^2.$$

To avoid square roots entirely, we work with *squared distance*.

¹based on field knowledge and empirical evidences

Definition 1 (Squared distance) For points $p = (x_1, y_1)$ and $q = (x_2, y_2)$ in $\mathbb{R}^2$, define

$$\text{dist2}(p, q) = (x_1 - x_2)^2 + (y_1 - y_2)^2.$$

Theorem 1 (Pythagoras on the axes) For all $a, b \in \mathbb{R}$,

$$\text{dist2}((a, 0), (0, b)) = a^2 + b^2.$$

3 Formalization in Rocq/Coq

We now present a complete mechanized proof of Theorem 1. The proof uses only the standard real numbers library and the `ring` tactic for normalization [3].

3.1 Code Listing

Listing 1 Formalization of the Pythagorean Theorem (squared form) in Rocq/Coq.

```
From Coq Require Import Reals.
From Coq Require Import Ring.

Open Scope R_scope.

Definition Point : Type := (R * R)%type.

Definition dist2 (p q : Point) : R :=
  let '(x1, y1) := p in
  let '(x2, y2) := q in
  (x1 - x2) * (x1 - x2) + (y1 - y2) * (y1 - y2).

Definition sq (x : R) : R := x * x.

Theorem pythagore_axes (a b : R) :
  dist2 (a, 0) (0, b) = sq a + sq b.
Proof.
  unfold dist2, sq.
  simpl.
  ring.
Qed.
```

4 Results

”No more goals.”

5 Conclusion

We provided a complete formal proof of a classroom theorem that almost no one asked to see formalized. In doing so, we confirm that the Pythagorean theorem is compatible with mechanized reasoning, and that parentheses can be trusted more than diagrams.

6 Contributors

- Andreas Mulard: *Semicolon expert, responsible for spelling mistakes*
- Yacine Hbada: *Expert in unknown unknowns*
- Léonie Gaztelumendi: *Professional eater*
- Thomas Mairesse: *Specialist in information technology and the space programs of archaic Greek societies*
- Romain Brouard: *Particle accelerator*
- Madhav Cherupilil Sajeed: *Cheeseburger taster*
- Alexis Pocquet: *One day, I tried to eat grass*
- Fernando Leal: *Assistant to the regional manager*
- Florian Chivé: *Z-letter typing assistant*
- Lyam Goux: *Official zumba dancer of the research team*
- Tran Décaudin: *The basic none guy*
- Tarkus the Armadillo-Tank: *Senior autistic spirit conduit*
 - The spirits have autism, not me.
 - Well, *I do* have autism, but the title refers to the spirits, who indeed have autism.
 - Actually, maybe having autism helps with the job.
 - We do not have a non-autistic spirit problem in the lab, for obvious reasons.
 - I would be able to deal with non-autistic spirits, but my employers wanted an accurate job title.
 - Because they have autism.
 - I hope they do not die on the job.
 - Well... I hope they do not die *anywhere*.
 - Did you know that in medieval fortresses, spiral staircases usually go up clockwise? This gave defenders an advantage in duels against attackers, since most people are right-handed. That is a cool fact.
 - You lost the Game.
 - An essential role, because otherwise you can hear them annoyingly whisper “Chooo chooooooooo,” which is very distracting.
- Maike Päsler: *Creator of pie charts for color-naming data in SPSS*
- Gabriel Berthel: *Peer-review*
- Max Lemoine: *Yes*

Declarations

- Funding: Not applicable.
- Conflict of interest/Competing interests: The authors declare no competing interest.
- Ethics approval and consent to participate: Not applicable.
- Consent for publication: Not applicable.
- Data availability: Not applicable.
- Materials availability: Not applicable.
- Code availability: Provided in Listing [1](#).

References

- [1] Chambers, P.: Teaching pythagoras' theorem. *Mathematics in School* **28**(4), 22–24 (1999)
- [2] Geuvers, H.: Proof assistants: History, ideas and future. *Sadhana* **34**(1), 3–25 (2009) <https://doi.org/10.1007/s12046-009-0001-5>
- [3] Huet, G., Kahn, G., Paulin-Mohring, C.: *The coq proof assistant a tutorial* (2010)